ASMI BASNET

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PROFESSIONAL SUMMARY

Machine Learning Engineer and Applied AI Specialist (Master of Science in Analytics candidate, Saint Louis University; B.E. in Electronics & Communication) with 3+ years of experience engineering production predictive ML models, generative AI architectures, and intelligent clinical applications. Proven expertise analyzing complex structured and unstructured healthcare datasets (34,000+ EHR records, clinical notes, imaging), training ensemble classifiers (XGBoost, Random Forest, PyTorch), and deploying Explainable AI (SHAP TreeExplainer) to ensure clinical transparency. Skilled in architecting privacy-first, offline retrieval-augmented generation (RAG) suites with automated citation verification (PolicyGuard), time-series forecasting (Prophet + LSTM), and modular SQL across enterprise databases. Florida resident eager to deliver high-impact, compliant, and reliable AI/ML solutions for Jackson Health System.

TECHNICAL SKILLS & CORE COMPETENCIES

Machine Learning & Modeling: Supervised & Unsupervised ML (Scikit-Learn, XGBoost, LightGBM, Random Forest, Lasso), Deep Learning (PyTorch, Torchvision, CNNs, LSTM), Explainable AI (SHAP TreeExplainer), Time-Series Forecasting (Prophet, ARIMA), K-Fold Cross-Validation, Hyperparameter Optimization (Hyperopt).
Generative AI & LLMs: Retrieval-Augmented Generation (RAG), Local LLM Inference (Ollama: Qwen 2.5, Llama 3, Phi-3), OpenAI GPT-4o API, Anthropic Claude API (Prompt Caching & Tools), Vector Databases (ChromaDB, FAISS), Citation Verification, Clinical Guardrails, Agentic Workflows (Agno, WebMCP).
Programming & Infrastructure: Python (FastAPI, Flask, Pandas, NumPy, Scikit-Learn, PyTorch, SQLAlchemy), SQL (Snowflake, PostgreSQL, DuckDB, SQL Server, BigQuery: CTEs, Window Functions), C/C++, TypeScript, Linux/Bash CLI, Git / GitHub, Docker, Docker Compose, RESTful APIs, Webhooks.
Healthcare Data & Governance: EHR Data Modeling, Clinical Informatics (Synthea EHR), DICOM / Medical Imaging, Automated Data Quality Assurance (IV&V), Schema Drift Detection, HIPAA / FERPA Privacy Compliance, Audit Logging, Business Intelligence (Power BI, DAX, Tableau).

PROFESSIONAL EXPERIENCE

Saint Louis University — **St. Louis, MO**
Graduate Machine Learning & AI Researcher (SPS-Corps) — *August 2025 – Present*

- Architected and deployed an air-gapped, privacy-first offline RAG compliance suite in Python (FastAPI, ChromaDB, Ollama) that runs local quantized LLMs to process sensitive policy and clinical guidelines with zero external data exposure.
- Engineered a chain-of-thought verification engine evaluating retrieved context sentence-by-sentence, flagging contradictory requirements (FLAG_CONFLICT), and requesting missing context (CLARIFY) to automate governance audits.
- Built a hybrid time-series forecasting engine combining Facebook Prophet with PyTorch LSTM neural networks and automated anomaly detection (flagging variances beyond $\pm 2.5\sigma$) to deliver operational demand and resource predictions.
- Optimized modular SQL queries across 8+ operational databases using complex CTEs and analytical window functions (ROW_NUMBER, DENSE_RANK), standardizing data dictionaries and reducing query execution latency by 40%.
- Implemented automated data quality verification suites (IV&V) enforcing referential integrity, schema drift detection, and rigorous data governance standards aligned with institutional and privacy compliance frameworks.
- Collaborated directly with cross-functional leadership in Agile sprints to translate operational requirements into technical specifications, authoring runbooks and conducting user training workshops for 25+ staff members.

Saint Louis University — **St. Louis, MO**
Data & AI Solutions Analyst Intern — *June 2024 – August 2024*

- Engineered automated data profiling, cleaning, and validation pipelines in Python (Pandas) and SQL across 10,000+ transactional records, establishing automated webhook alerts that instantly flag schema drift and data discrepancies.
- Built and deployed 6 interactive Power BI and Tableau executive dashboards backed by star-schema data models, tracking 15+ operational KPIs to enable leadership to monitor resource allocation, productivity, and project milestones.
- Enforced strict institutional data governance and privacy rules, building row-level data quality validation checks across reporting layers.

Entegra Sources Pvt. Ltd. (Mayahold Group) — **Remote / Kathmandu, Nepal**
Systems & Data Analyst — *December 2022 – July 2023*

- Extracted, cleansed, and normalized multi-source operational and telemetry datasets (5,000+ records) into standardized 3NF relational schemas using Python (Pandas, SQLAlchemy) and SQL, guaranteeing complete data fidelity.
- Validated algorithmic logic and automated processing scripts in Python and C/C++, implementing robust exception handling, boundary tests, and audit logging to maintain 99.8% pipeline reliability.

FEATURED CLINICAL AI & MACHINE LEARNING PROJECTS

CareGap — Clinical Decision Support & Population Health Analytics — **Python, DuckDB, Scikit-Learn, XGBoost, SHAP, FAISS, Streamlit**

- Developed a full-stack clinical decision platform querying 34,000+ longitudinal synthetic EHR patient records with sub-second latency using DuckDB columnar storage, training an ensemble classifier (XGBoost, Random Forest, Lasso) for patient risk stratification.
- Integrated SHAP TreeExplainer for per-patient transparent risk attribution alongside a dual-tier RAG clinical assistant (FAISS + local Ollama with failover to Gemini) featuring strict guardrails that block unauthorized dosage recommendations.

PolicyGuard — Enterprise Offline RAG & Compliance Intelligence Suite — **Python, Ollama (Qwen 2.5 / Llama 3), ChromaDB, FastAPI, Docker**

- Engineered an air-gapped, privacy-first RAG application running entirely on local quantized LLMs via Ollama to analyze confidential enterprise documentation and clinical guidelines with zero third-party API exposure and zero data leakage.
- Implemented automated chain-of-thought citation verification and dynamic cosine similarity guardrails, evaluating retrieved context sentence-by-sentence to detect contradictory clauses (FLAG_CONFLICT) and eliminate manual review overhead.

WashU Clinical Diabetes Risk Prediction Datathon — **Python, Scikit-Learn, XGBoost, LightGBM, Hyperopt**

- Engineered an end-to-end ML pipeline on 62,000+ clinical EHR records to predict uncontrolled HbA1c, engineering visit-frequency, medication-count, and glycemic-volatility features.
- Benchmarked Logistic Regression, Random Forest, LightGBM, and XGBoost using stratified K-fold cross-validation and hyperparameter tuning, optimizing ROC-AUC, PR-AUC, and F1 to minimize clinical false negatives.

EDUCATION & CERTIFICATIONS

Saint Louis University | Master of Science in Analytics (STEM) — **St. Louis, MO | Expected May 2026**
Tribhuvan University | Bachelor of Engineering in Electronics & Communication (STEM) — **Kathmandu, Nepal | Conferred 2023**

Certifications: AI Engineer for Data Scientists Associate (DataCamp: PyTorch Deep Learning & Scikit-Learn ML Pipelines) · Building with Claude & Anthropic API (Prompt Caching & Tools) · Microsoft Certified: Power BI Data Analyst Associate (PL-300) · Project Management & Agile Certification.